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Magnetic fabrics and paleomagnetism of the Storsjön–Edsbyn deformation zone, central Sweden

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Abstract

A regional deformation zone, separating tectonomagmatic blocks in the central part of the Fennoscandian Shield, has been studied using anisotropy of magnetic susceptibility and paleomagnetism in order to define the tectonic character and age of the zone and to put the deformation into a plate tectonic context. The deformation zone (the Storsjön–Edsbyn Deformation Zone; SEDZ) separates the Rätan granite of the Trans Scandinavian Igneous Belt (TIB) and the southern Svecofennian subprovince in the southwest from older Svecofennian granitoids of the central Svecofennian subprovince in the northeast. The SEDZ is the most prominent deformation zone in a regional pattern of NW–SE trending shear zones. On basis of AMS data the width of the zone, which cuts through 1.7–1.9 Ga old rocks, is estimated at 10–12 km in its central–southern part and 20–25 km in its northern part. The orientations of the susceptibility axes vary systematically across the zone, in a way that suggests that the SEDZ is a pure shear dominated transpression zone caused by a compressive stress from the present SSW. Paleomagnetic data from a dolerite sill (Plat = 10°, Plon = 166°) and from a dyke (Plat = 25°, Plon = 190°) that cut the SEDZ indicate that no plastic deformation has occurred along the zone during the past 1.25 Ga, and probably not for the past 1.5–1.6 Ga. AMS data do not point to any deformation of the 1.70 Ga old Rätan granite, which further limits the time span of deformation. The main phase of deformation most likely took place between ~1.80 and 1.70 Ga ago. The geological evolution of this part of the Fennoscandian Shield at the time between 1.80 and 1.70 Ga ago is very complex and in part still enigmatic. This time period involves magmatism of the TIB, which is related to eastward subduction. It also comprises the beginning of the Gothian orogeny and the development of a regional system of transpressive deformation zones. The compressive stress field from the present SSW indicated in this study is interpreted to be related to an initial stage of plate collision between the southwestern Scandinavian subprovince and older parts of the Fennoscandian Shield. This collision resulted in a system of regional NW–SE trending intra-plate deformation zones, one of which is the SEDZ © 2001 Elsevier Science B.V. All rights reserved.

Keywords: Magnetic anisotropy of susceptibility; Paleomagnetism; Proterozoic deformation zone; Sweden; Transpression

1. Introduction

The Fennoscandian Shield can be divided into a number of tectonomagmatic blocks (Gaál and

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Gorbatschev, 1987; Elming et al., 1993), that are separated by more or less well defined tectonic zones (Fig. 1). The tectonic blocks reflect a geochronological zonation of the shield and a better knowledge of the character of the deformation zones that separate them is vital for an understanding of the geological evolution of Fennoscandia.

The Svecofennian Domain of Paleoproterozoic rocks dominates the central part of the Baltic Shield. Along the western margin of the Svecofennian domain an almost N–S trending area of granitoid rocks, the Transscandinavian Igneous Belt (TIB), was formed between ~1.81 and 1.66 Ga ago (Åhäll and Larson, 2000). The Storsjön–Edsbyn deformation zone (SEDZ) is a regional tectonic zone that separates the TIB and the southern Svecofennian subprovince from the central Svecofennian subprovince (Fig. 1). The zone trends in a NNW direction and extends for a

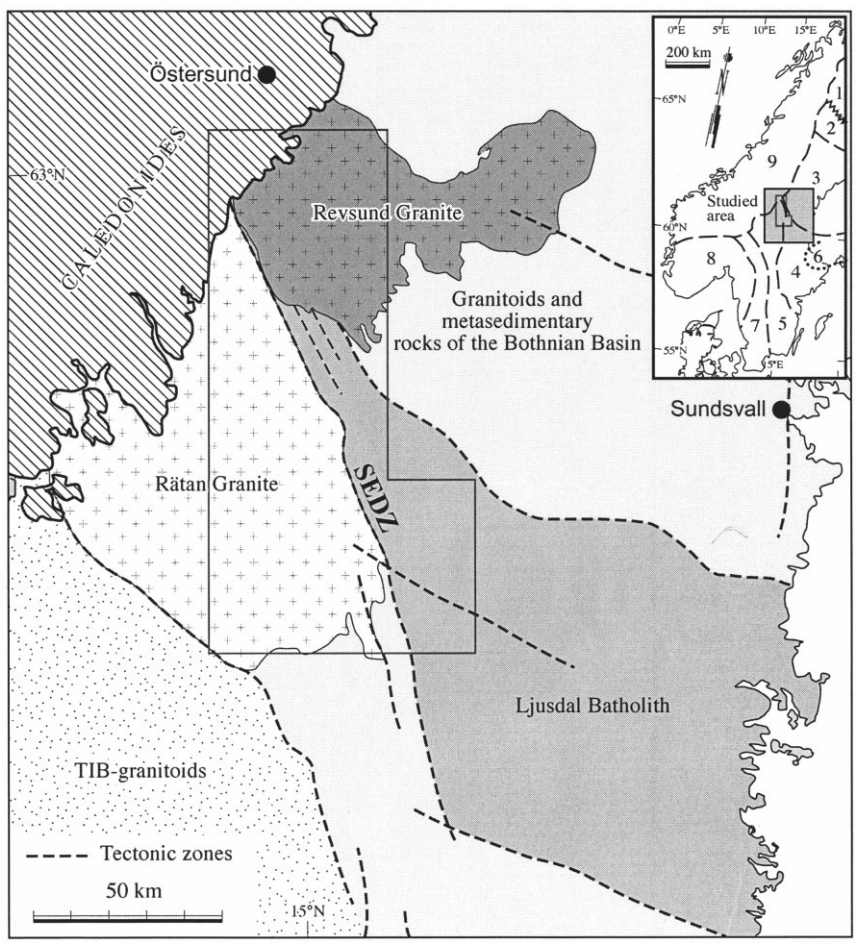


Fig. 1. Generalized geological map showing the tectonic setting and major deformation zones of the central parts of the Fennoscandian Shield, Sweden (re-drawn from the National Atlas of Sweden, 1995 and inferred from Högdahl and Sjöström, 1999; Bergman and Sjöström, 1994). The frame in the center of the figure indicates the location of Fig. 2. The inset diagram in the upper right corner shows the location of the study area with respect to the Fennoscandian Shield. It also shows a tectonomagmatic block division of Fennoscandia (modified after Elming et al., 1993). 1 = Northern Svecofennian subprovince, 2 = Skellefte belt, 3 = Central Svecofennian subprovince, 4 = Southern Svecofennian subprovince, 5 = Transscandinavian igneous belt (TIB), 6 = Rapakivi granites, 7 = Protogine zone, 8 = Southwest Scandinavian subprovince, 9 = Caledonides.

distance of ~200 km, and it forms the most prominent part of a regional system of NW–SE trending deformation zones in the central part of the Baltic Shield (Bergman and Sjöström, 1994) (Fig. 1). These zones are believed to have developed during the time period 1.65–1.82 Ga ago (Bergman and Sjöström, 1994; Högdahl et al., 1996; Högdahl and Sjöström, 1999).

The beginning of the Gothian orogeny (~1.75–1.55 Ga; Åhäll and Larson, 2000) coincide in time with the magmatism of the TIB. The geological evolution of the TIB is fairly well documented (e.g. Ahl et al., 1999; Åhäll and Larson, 2000) whereas the origin and mechanisms of the Gothian evolution are rather poorly understood (Åhäll and Larson, 2000). With this study we want to contribute to a better understanding of the tectonic evolution related to the Gothian orogeny, and this we do by investigating the SEDZ. Some questions to be answered are; what is the type of deformation? Can we define the paleo stress field? Can we estimate the age of deformation? Is it possible to put the development of the SEDZ into a plate tectonic context? To answer these questions we have used anisotropy of magnetic susceptibility (AMS) and paleomagnetism in a study concentrated in the northern part of the zone, where it marks the boundary between the Rätan granite of the TIB in the west and the Ljusdal batholith and Revsund granite of the central Svecofennian subprovince in the east (Fig. 1). A previously presented tectonic model of the zone (Bergman and Sjöström, 1994) is also tested.

2. Geological setting

Three generations of granite–granodiorite massifs dominate the geology of the region (Fig. 1). In the southeast there is the strongly foliated Ljusdal batholith, a granodiorite that is ~1.84 Ga old (U–Pb, Delin, 1993). The Svecofennian orogeny, with two phases of folding plus a partial migmatization (Lundqvist et al., 1990), re-worked the batholith and gave rise to a regional schistosity with mainly shallow dipping planes of foliation. In the west there is the magnetite and hematite bearing 1.70 Ga old Rätan granite (U–Pb, Delin, 1996) that is related to the Transscandinavian Igneous Belt (TIB). The rectangular shape of the granite can be fairly well defined from aeromagnetic data and its eastern boundary coincides with the SEDZ for ~100 km. The grayish and coarsely porphyritic ~1.80 Ga old Revsund granitoid batholith (U–Pb, Claesson and Lundqvist, 1995) is emplaced in the northeast and two minor intrusions of Revsund type granites (1.80 Ga old, U–Pb; Delin, 1996) are emplaced southwest of the SEDZ. In contrast with the Ljusdal batholith, the Rätan and Revsund granitoids have not undergone regional metamorphism (Delin and Aaro, 1992; Lundqvist et al., 1990). Dolerite sills and dykes intrude large areas of the Rätan and Revsund granitoids in the northern part of the study area. They are thought to belong to a major formation of dolerites that is found in the central and northern parts of Sweden and also in the south western parts of Finland. Radiometric dating (U–Pb) in the archipelago between Sweden and Finland gives an age of 1.27 Ga of these rocks (Suominen, 1991).

3. The Storsjön–Edsbyn deformation zone (SEDZ)

The system of NW–SE trending shear zones, to which the SEDZ belongs, characterizes the structural evolution of the Svecofennian Domain (Fig. 1). Bergman and Sjöström (1994) distinguish four groups of shear zones in the area of the SEDZ, (I) plastic, syn metamorphic, (II) mostly plastic, retrograde, (III) brittle-plastic and (IV) brittle, post-metamorphic. Types (II) and (III) are believed to be typical for the SEDZ whereas type (I) is connected to the regional folding and thought to be older. Type (IV) is the youngest in this classification (Bergman and Sjöström, 1994). The age of the type (II) and (III) shear zones is by Bergman and Sjöström (1994) believed to be ~1.6–1.7 Ga, or possibly younger. However, U–Pb dating of syn tectonic titanites in SEDZ-mylonites (area B; Fig. 2) indicates an age of the deformation of ~1.80 Ga (Högdahl et al., 1996). A number of shear zones situated east and southeast of the

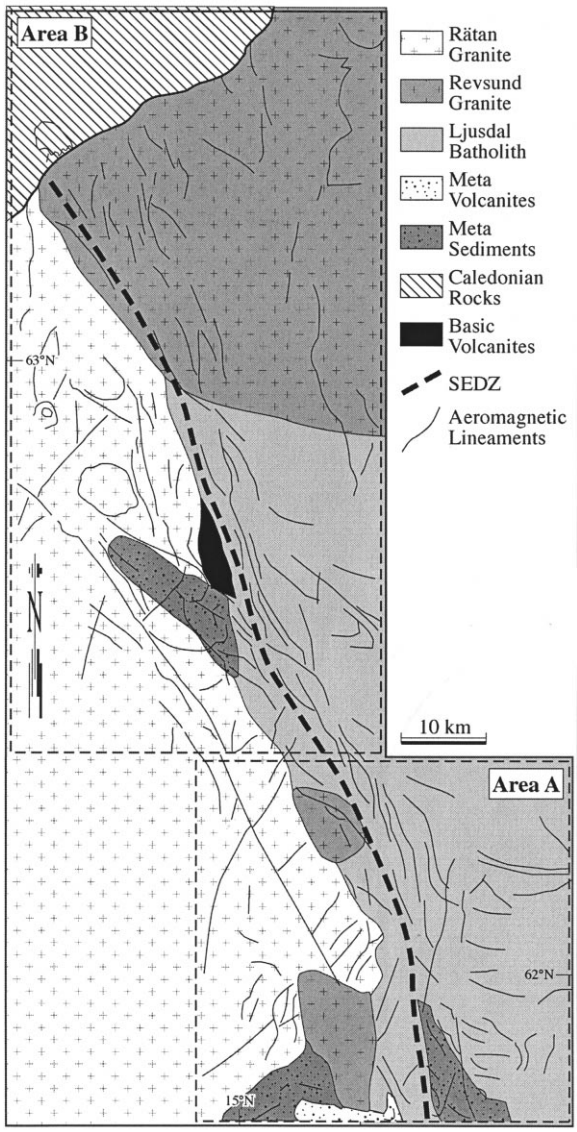


Fig. 2. Generalized geological map showing the study area in central Sweden. The map is based on aeromagnetic data (Geological Survey of Sweden) and simplified from Lundegårdh et al. (1984), Lundqvist et al. (1990), Delin and Aaro (1992, 1994).

SEDZ, suggested being related to a pattern of NW–SE trending zones, are dated at ~1.80–1.82 Ga (U–Pb, Högdahl and Sjöström, 1999).

A strain analysis indicates a non-coaxial deformation of flattening type with sub vertical maximum strain and a NW–SE striking XY plane of the strain ellipsoid (Bergman and Sjöström, 1994). The plastic mylonites (type II) in the northern part of the SEDZ (area B in Fig. 2) show dip slip lineation, whereas the brittle plastic shear zones (type III) have sub-horizontal slip lineation, which indicates a strike slip motion. Most of the shear zones have sub-horizontal, NE–SW striking foliation poles. Bergman and Sjöström (1994) conclude that the SEDZ is a transpression zone with a dextral strike slip component of c. 16km. Bergman and Sjöström (1994) also suggest that there is a causal relationship between the intrusion of the Rätan granite and the SEDZ since the magma would provide the necessary heat for the deformation and the zone broadly follows the eastern boundary of the granite. The zone however continuous ~50 km south of the southern boundary of the Rätan granite, which indicates that the granitoid hardly is solely responsible for the development of the SEDZ.

4. Methods and sampling sites

Anisotropy of magnetic susceptibility (AMS) was used to measure the petrofabric of rocks (e.g., Hrouda, 1982). Basically, the AMS of a rock is described by a second-order tensor. This tensor can be visualized by an ellipsoid having three principal axes of maximum, intermediate and minimum susceptibility (k_{max}, k_{int} and k_{min}, respectively). For multi-domain magnetite the k_{max} axis is parallel to the long axis of an ellipsoidal grain and for most rocks the k_{max} corresponds to the mineral lineation, whereas the k_{min} axis corresponds to the pole to the plane of foliation (Tarling and Hrouda, 1993). The degree of anisotropy is characterized by $P = k_{max}/k_{min}$ and the shape of the anisotropy ellipsoid is defined by the shape parameter T (Jelínek, 1981). If $-1 < T < 0$ the ellipsoid is prolate (cigar-shaped) and if $0 < T < 1$ the ellipsoid is oblate (disc-shaped); if $T = 0$ the ellipsoid is neutral. The AMS and temperature variation of magnetic susceptibility of rocks were measured with a Kappabridge KLY-3 and a CS-3 apparatus, respectively (Geofyzika Brno). The statistical analysis of the AMS data was performed using the technique of Jelínek (1978).

Direction and intensity of the remanent magnetization was measured with a spinner magnetometer JR-5 (Geofyzika Brno) and a DC-SQUID magnetometer (2G-Enterprises). Alternating field demagnetization was performed with coils incorporated in the DC-SQUID system and thermal demagnetization was performed using a Schonstedt demagnetizer TSD-1. Remanent magnetization directions were analyzed using Principle Component Analysis (Kirschvink, 1980) and means were calculated using the statistics of Fisher (1953).

The samples were collected with a portable field drill and orientated with a magnetic and a sun compass. In area A (Fig. 2) 135 drill cores were collected at 16 sites distributed along a ~33 km long trajectory (Fig. 3). Twelve of these sites are from the Ljusdal batholith (KA01–KA12) and four are from the Revsund type granite (KA13–KA16). In the northern part of area B (Fig. 2) 234 samples were collected at 20 sites. Seven sites are situated in the Revsund granite (ST01–ST07), five are from the Rätan granite (ST11–ST15) and five are dolerites (ST16–ST18, ST21–ST22) (Fig. 4).

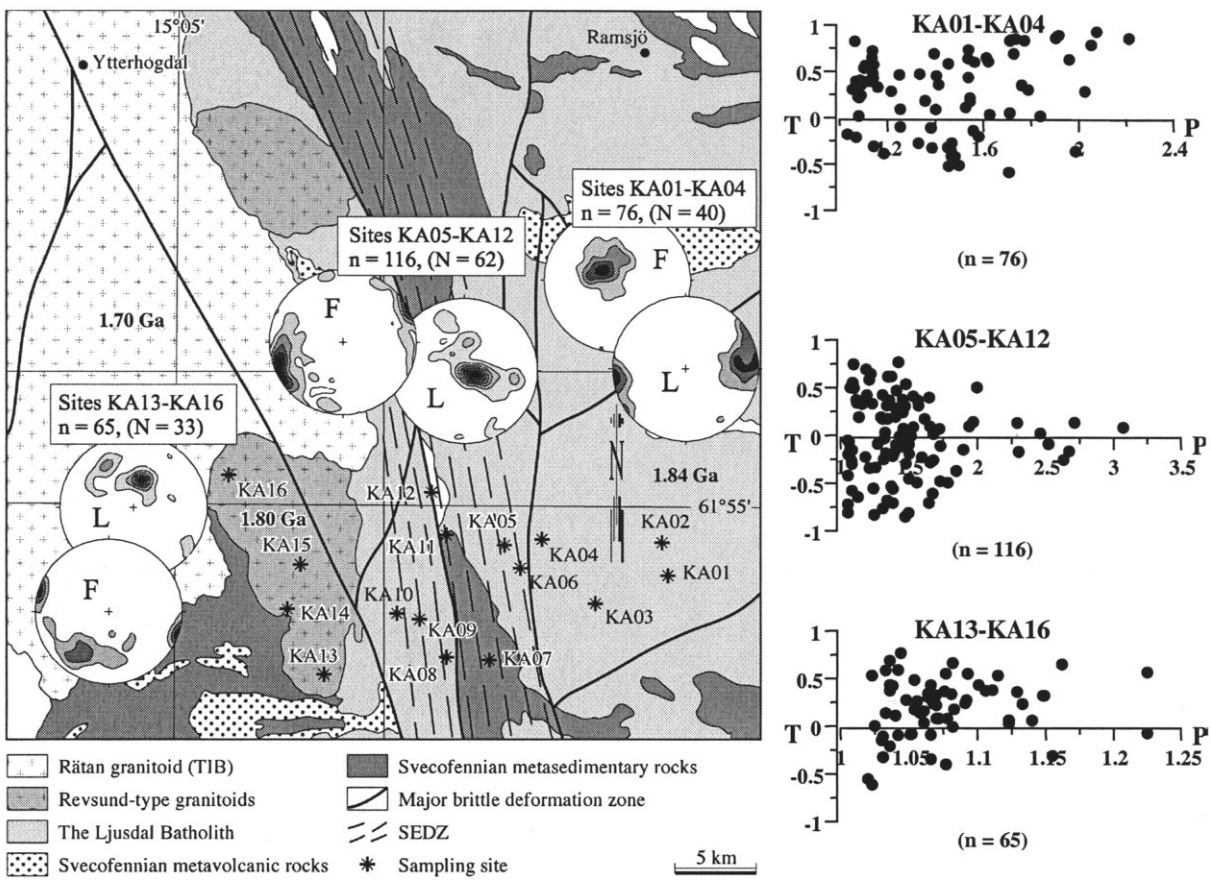


Fig. 3. Geological map of the southern part of the SEDZ (area A in Fig. 2) simplified from Delin and Aaro (1992). Locations of the studied sites are shown together with contour plots of directions of minimum (foliation poles — F) and maximum (lineation — L) susceptibility axes in stereographic projections for all sites separated in three groups. The contours are calculated with reference to the number of specimens (n), and N denotes the number of samples. P–T diagrams (Jelínek, 1981) are also shown for the three groups of sites.

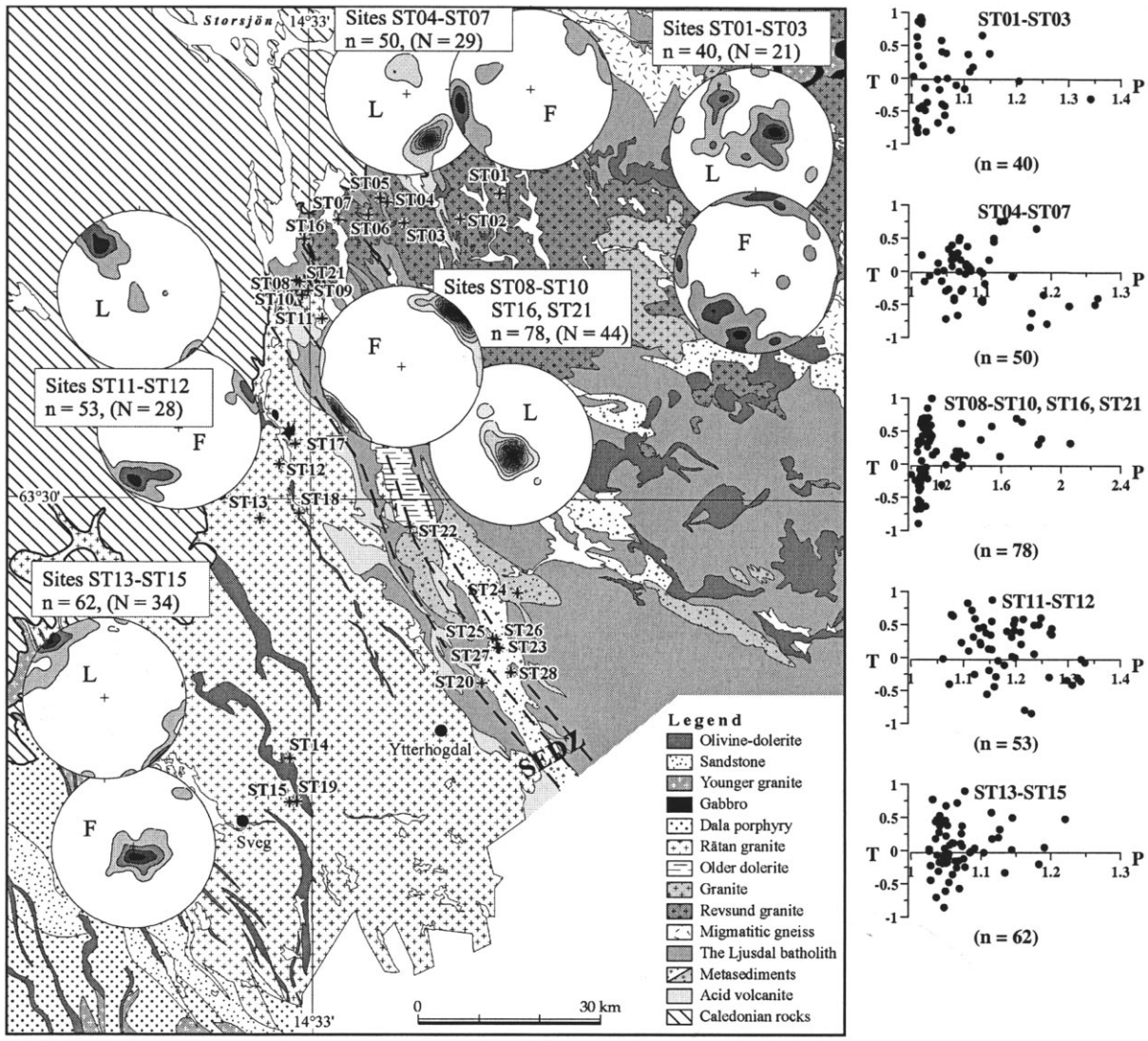


Fig. 4. Geological map of the northern part of the SEDZ (area B in Fig. 2), simplified from Lundegårdh et al. (1984), Lundqvist et al., (1990). Locations of the studied sites are shown together with contour plots of directions of minimum (foliation poles — F) and maximum (lineation — L) susceptibility axes in stereographic projections for five groups of sites. The contours are calculated with reference to the number of specimens (n), and N denotes the number of samples. P–T diagrams (Jelínek, 1981) are also shown for these groups of sites.

The remaining sites are from two mylonites (ST08–ST09) and one augen gneiss granite (ST10). In the southern part of area B 89 drill cores were collected at eight different localities, including five dolerites (ST19–ST20, ST22, ST24, ST26), one granite site (ST25) and two gneiss rocks (ST27–ST28).

5. Results

5.1. Petrography

The rock in the four sites KA01–KA04 of the Ljusdal batholith (Fig. 3) is a medium- to fine-grained granodiorite that possesses a sub-horizon-

tal biotite foliation. It carries crystals of hornblende, biotite, plagioclase and quartz, without signs of high-grade deformation. Apatite grains are often associated with biotite. Magnetite and ilmenite lamellae dominate the opaque mineral fraction.

In the central part of the SEDZ (sites KA05–KA12, Fig. 3) the granodiorite has suffered from variable degrees of chlorite alteration. Zones of high-grade deformation are indicated by steeply dipping foliation defined by biotite and elongated quartz grains.

The Revsund type granite west of the SEDZ (sites KA13–KA16, Fig. 3) carries lamellae of ilmenite that are frequently seen in between flakes of biotite. Magnetite crystals appear unaltered, and they often surround the grains of ilmenite. A few pyrite grains have borders of hematite, or iron oxyhydroxide, which may indicate chemical alteration.

The Revsund granite in the northeast (Fig. 1) is coarse-grained and dominated by megacrysts of plagioclase, microcline, quartz and biotite. The three easternmost localities (ST01–ST03, Fig. 4) are unaltered, apart from a minor chlorite alteration of the rock at site ST01. West of these localities, at sites ST04–ST07, the granite has a steeply dipping biotite foliation trending NNW–SSE. Gneisses, acid volcanics and banded plastic mylonites (sites ST08–ST10, ST1 and ST21) define the central part of the SEDZ with distinctly developed foliations that trend NW–SE and dip sub-vertically. The microscope analysis of a sample from site ST10 (augen gneiss granite) shows that some magnetite grains are deformed and have an overgrowth of epidote, while others are cubic in shape and appear unaltered. This indicates two generations of magnetite, of which the unaltered fraction most likely is post-deformational.

The Rätan granite (sites ST11–ST15) only shows signs of low-grade deformation indicated by altered mica crystals and fractures filled with fine-grained quartz and feldspars.

5.2. AMS

The aeromagnetic anomaly pattern of the Ljusdal batholith shows E–W trending anomalies in the east that rotate clockwise to a NNW–SSE direction when approaching the SEDZ. The rotation suggests a dextral shear sense of the zone. The AMS data of the batholith, east of the SEDZ (sites KA01–KA04, area A), show sub-horizontal planes of magnetic foliation and E–W striking magnetic lineation (Fig. 3). The degree of anisotropy (P) ranges from almost isotropic to more than 100% anisotropy and most samples have oblate shaped AMS ellipsoids (Fig. 3). Inside the deformation zone (sites KA05–KA12) most planes of magnetic foliation trend in a NNW–SSE direction and dip sub-vertically. Maximum and intermediate susceptibility axes are scattered or define a girdle pattern (Fig. 5A), which results in prolate as well as oblate shaped anisotropy ellipsoids. In the aeromagnetic data this fabric is cut by a number of N–S trending low magnetic lineaments. The sites at these lineaments, or close to them, have clustered anisotropy axes with a sub-vertical lineation (Fig. 5B) and mainly prolate ellipsoids. Rock magnetic data indicate that magnetite is the main magnetic carrier in the Ljusdal batholith both inside and outside of the SEDZ (6A, 7B). However, some minor amounts of hematite also contribute to the susceptibility (site KA09), and at one site (KA03) the fabric is completely governed by paramagnetic minerals (Fig. 6A). The bulk susceptibility is similar between the sites KA01–KA04 (outside the SEDZ) and the sites KA05–KA12 (inside the SEDZ) (Fig. 7B).

The AMS data of the Revsund type granite situated west of the SEDZ (sites KA13–KA16) do not indicate a coherent fabric. However, most samples have oblate shaped anisotropy ellipsoids (Fig. 3). The poles to the magnetic foliation form two clusters, one that trends SSW with a moderate dip, and the other that is sub-horizontal and trends ESE–WNW (Fig. 3). The magnetic lineations cluster around a strike direction to the north with a moderate dip. At site KA16 we can define a preferred direction of the feldspars that trends NW–SE, which is parallel to the magnetic foliation plane.

In area B (Fig. 2) the magnetic fabric of the Revsund granite at the localities ST01–ST03 is rather poorly developed. The minimum principal susceptibility axes define sub-vertical planes trend-

ing in various directions and the lineation varies from sub-horizontal to sub-vertical. Closer to the SEDZ, at the sites ST04–ST07, the magnetic foliation has a NNW–SSE trending direction and a sub-vertical dip, which is parallel to the biotite foliation observed in the field (Fig. 4). The lineation mainly strikes SSE and dips moderately, indicating a dominant strike-slip component of deformation. Paramagnetic minerals (most likely the observed biotite) is the main carrier of magnetic susceptibility of the Revsund granite, except at site ST05, which is a recrystallized mylonite that carries magnetite (6B, 7A). The degree of anisotropy is below 10% at the unaltered sites (ST01–ST03) and only slightly higher at the deformed sites (ST04–ST07).

In the mylonites and the gneissic rocks in the center of the zone (sites ST08–ST10, ST16 and ST21, Fig. 4) the poles to the magnetic foliation are tightly clustered in the NE–SW direction, with a sub-horizontal dip and the lineation clusters sub-vertically. The fabric most likely expresses a dominant NE–SW directed pure shear component that results in a sub-vertical maximum finite strain.

The degree of anisotropy in specimens varies from isotropic (site ST16) to $P = 2.06$ at site ST09, which most likely reflects differences in magnetic mineralogy; the mean susceptibility $K_{mean} = 0.00057 \pm 0.00005$ SI (site ST16), $K_{mean} = 0.004 \pm 0.008$ SI (site ST09). A linear relation between P and K_{mean}, defined at site ST10, supports this (Fig. 6D).

Crossing the SEDZ westward and moving into the Rätan granite, the magnetic foliation planes of the rock at the sites ST11–ST12 are parallel to the general direction of the zone (Fig. 4). This orientation is also parallel to the eastern boundary of the Rätan granite. Further to the SW, at the sites ST13–ST15, the magnetic foliation has flipped $\sim 90°$ to a sub-horizontal orientation. The specimens are more oblate shaped and the degree of anisotropy is slightly higher at the sites ST11–ST12 than at ST13–ST15 (Fig. 4). This points to a difference in the stress conditions and a better grain alignment close to the contact than in the center of the granitoid. There is also a difference in bulk susceptibility between the sites at the boundary [(ST11–ST12) Kmean $= 0.015 \pm 0.008$SI] and in the center [(ST13–ST15) Kmean $= 0.029 \pm 0.011$SI], which indicates a possible enrichment of iron oxides toward the center of the pluton. The magnetic lineations of all the five sites in the Rätan granite (ST11–ST15) are

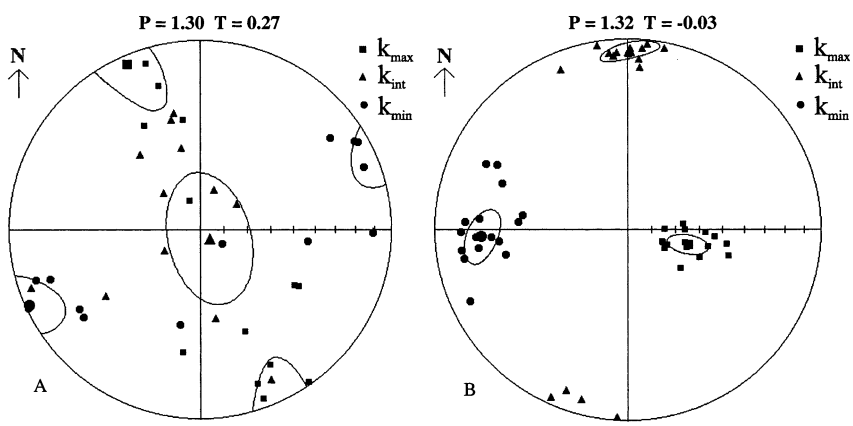


Fig. 5. Lower hemisphere equal area projection plots of AMS data showing examples of the NNW–SSE trending girdle distribution of maximum and intermediate anisotropy axes of a deformed rock (A; site KA09) that is cut by narrow deformation zones defined by tightly clustered anisotropy axes with sub-vertical lineation (B; site KA06). The ellipses are 95% confidence limits about each axis of the susceptibility ellipsoid.

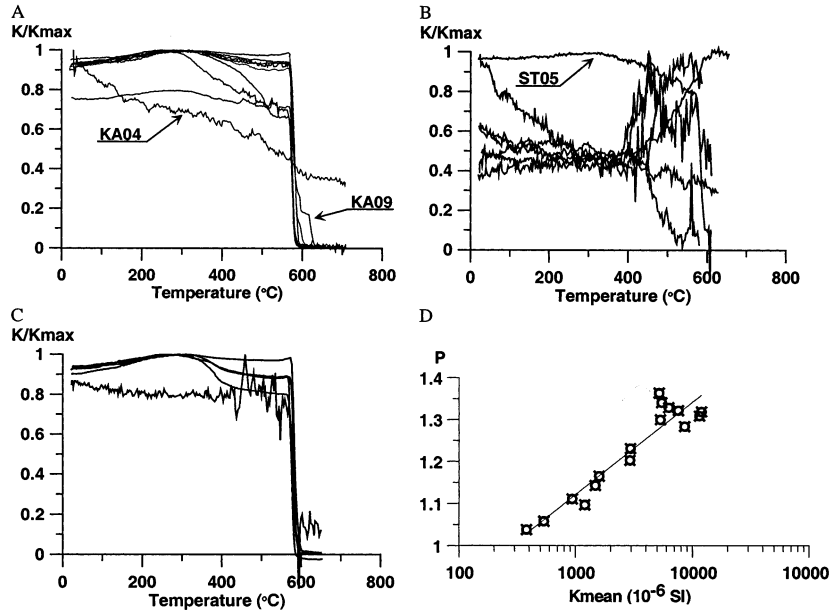


Fig. 6. Heating curves of the mean susceptibility for the Ljusdal batholith (A; eight samples), the Revsund granite (B; seven samples) and the Rätan granite (C; five samples). The degree of anisotropy (P) plotted with reference to the mean susceptibility for an augen gneiss granite, site ST10 ($n = 16$) (D).

sub-horizontal and strike NW–SE. This indicates that a large area of the northeastern part of the granite body has a sub-horizontal maximum finite strain striking NW–SE. Rockmagnetic data show that magnetite is the main carrier of the susceptibility in the Rätan granite (6C, 7C).

In the center of the SEDZ a dolerite sill truncates the deformation pattern of a plastic mylonite rock (site ST16, Fig. 4). The average degree of anisotropy of the sill is low ($P = 1.07$) and nothing indicates that the rock is deformed. Similar dolerite sills were sampled at four other localities outside the SEDZ (sites ST17–ST29) and close to its center (site ST20). The magnetic fabric of all five sites is consistent and they have sub-vertical minimum anisotropy axes and maximum and intermediate axes that define a girdle pattern in the plane of foliation (Fig. 9B). This fabric is considered characteristic for unaltered dolerite sills (Knight and Walker, 1988). A 20 m wide dolerite dyke at site 23 (Fig. 4) also seems to carry a primary magnetic fabric, even though an interpretation of aeromagnetic data indicates that the dyke is cut by a NNW–SSE trending brittle shear zone and dextrally displaced ~ 400 m. The average degree of anisotropy is low ($P = 1.03$) and the principal axes of the AMS ellipsoid correlate well with the dyke plane and its pole (Fig. 10B).

5.3. Paleomagnetism

Alternating field demagnetization of samples from the Ljusdal batholith (site KA02, Fig. 8A,B) shows that the rocks east of the shear zone area carry a stable remanent magnetization of decl. = 7° and incl. = 25° ($\alpha_{95} = 10°$, $k = 17$; Table 1). However, there is often an overlap in the coercivity spectra, which makes it difficult to isolate the characteristic magnetization. A great circle analysis (Bailey and Halls, 1984) shows that the intersections of the great circles (decl. = 358°, incl. = 30° and MAD = 12°; Fig. 8A) do not differ significantly from the mean direction defined from the vector diagrams (Table 1). This suggests that we have been able to isolate an ancient magnetization. The thermal data shows a wide range of unblocking temperatures (Fig. 8C) and the thermal mean direction is poorly constrained but it

indicates a direction similar to what is defined from the alternating field demagnetization (Table 1). The paleopole (Plat = 41°, Plon = 187°, $d_p = 6°$ and $d_m = 10°$) calculated from the mean remanence direction after AF demagnetization (Table 1) corresponds to a paleomagnetic age of 1.75–1.8 Ga (Fig. 11).

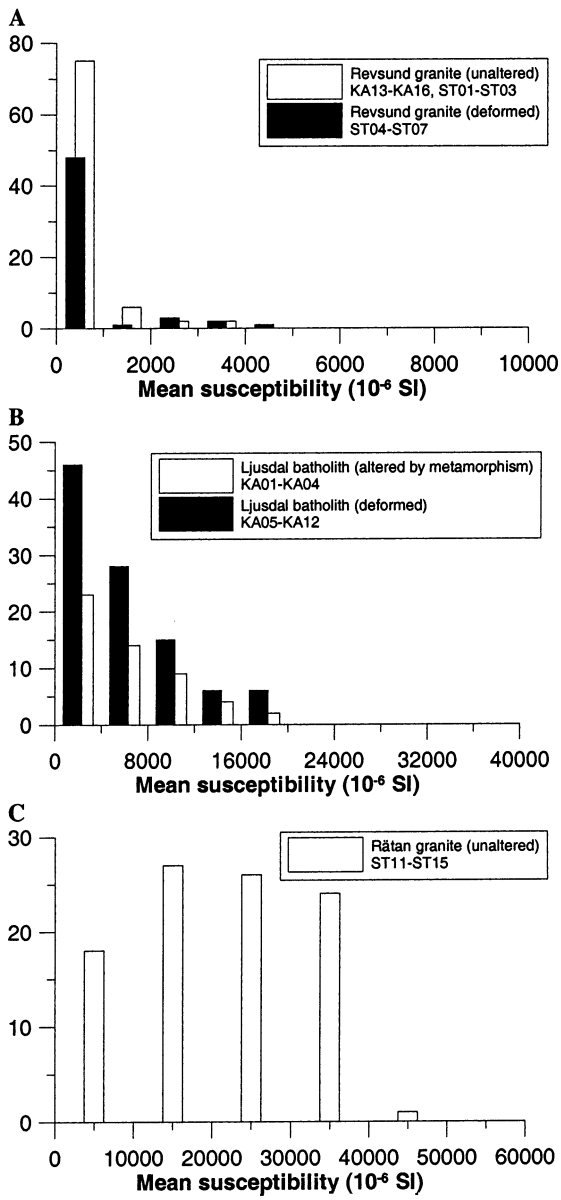


Fig. 7. The distribution of the mean susceptibility for the Revsund granite (A), the Ljusdal batholith (B) and the Rätan granite (C).

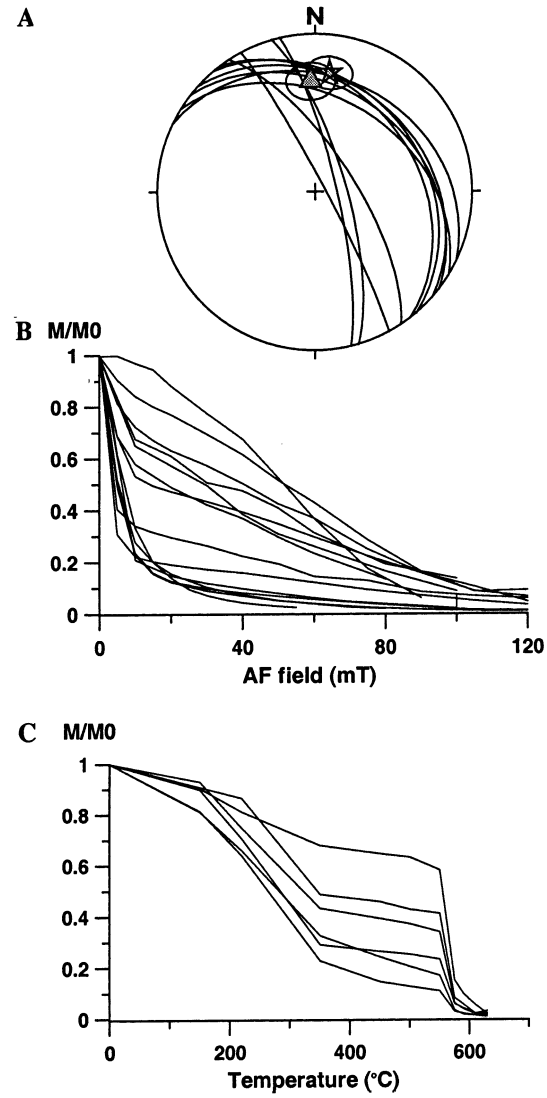


Fig. 8. Lower hemisphere equal area projection plot of great circles fitted to alternating field demagnetization sequences for 10 samples (maximum MAD accepted is 6.5°) from the Ljusdal batholith (site KA02) (A). The star denotes the mean direction defined from the vector analysis, the triangle shows the mean intersection direction of the great circles and the rings show the 95% cones of confidence of the two means. Alternating field and thermal demagnetization diagrams for 14 and six samples, respectively, from this site are shown in (B) and (C).

The dolerite sill that truncates the plastic deformation pattern in the zone at site ST16 carries a stable mean remanence direction after AF demagnetization (decl. = 29°, incl. = −26°, $\alpha_{95} = 6°$ and $k = 43$) that is likely to represent the primary magnetization of the rock (Table 1). Together with the other dolerite sills (ST17–ST20) a formation mean direction of decl. = 32° and incl. = −23° ($\alpha_{95} = 10°$, $k = 56$, 61 samples) is defined (Fig. 9A) and the calculated mean paleopole position (Plat = 12°, Plon = 162°, d_p = 6° and $d_m = 11°$) falls close to the 1.25 Ga reference pole on the APWP for the Fennoscandian Shield (Fig. 11). This indicates that the sills are of Postjotnian age. Thermal demagnetization and other rock magnetic data shows that magnetite is the main carrier of both the remanence and the susceptibility (Fig. 9C–E). The average median destructive field of 49 samples is 17 mT, and together with the fairly high Q-values (Fig. 9F) it indicates that grains of pseudo-single domain size carry the remanence (Dunlop and Özdemir, 1997).

The NNE trending dolerite dyke at site ST23 (Fig. 4) carries a stable remanence in high alternating fields and at high temperatures (Fig. 10C,D). A mean direction (decl. = 5°, incl. = −4°, $\alpha_{95} = 8°$ and $k = 26$; Table 1) is defined from the thermal data (Fig. 10A) and the paleopole (Plat = 25°, Plon = 190°, $d_p = 4°$ and $d_m = 8°$) calculated from this mean direction corresponds to two possible paleomagnetic age intervals of ~1.30–1.32 and ~1.5–1.6 Ga (Fig. 11). Neither of these two time intervals are represented in radiometric or paleomagnetic data from our study area. However, 1.5–1.6 Ga old northeast trending dykes with paleomagnetic directions similar to that defined at site ST23 have been identified in a dyke swarm in the central Svecofennian subprovince, ~100 km north of the SEDZ (Moakhar and Elming, 1998). It is possible that the dyke at site ST23 is related to this dyke swarm, and therefore should be placed in the 1.5–1.6 Ga time interval. The host rock of the dyke has a foliation that is parallel to the SEDZ, which, together with the close position of the sampling site relative to the center of the zone, indicates that the dyke post-dates all plastic deformation activity of the zone in this area.

Table 1
Paleomagnetic results[a]

Site	Met.	MAD (°)	$S/n/N$	Dec./Inc. (°)	α_{95} (°)	k	P_{lat}/P_{lon} (°)	d_p/d_m (°)
KA02	AF	5.8	1/15/16	7/25	10	17	41/187	6/10
	Th	3.1	1/13/14	19/31	30	3	43/171	19/33
ST16	AF	4.9	1/15/20	29/−26	6	43	10/166	4/6
	Th	1.3	1/4/4	29/−24	4	623	12/166	2/4
ST17	AF	5.5	1/3/4	42/−23	19	46	9/153	10/20
	Th	6.4	1/10/10	46/−18	8	43	10/149	4/8
ST18	AF	6.4	1/10/10	31/−19	8	38	14/163	4/8
	Th	1.3	1/3/3	31/−19	19	44	14/163	10/20
ST19	AF	4.2	1/16/16	28/−34	4	77	6/168	3/5
	Th	5.4	1/5/6	–	–	–	–	–
ST20	AF	6.8	1/4/5	26/−16	4	562	17/168	2/4
	Th	4.1	1/10/10	28/−16	7	54	17/166	4/7
ST23	AF	5.9	1/4/4	7/−19	27	12	18/188	15/29
	Th	4.7	1/14/17	5/−4	8	26	25/190	4/8
ST16–20	–	6.4	5/61/66	33/−23	10	56	12/162	6/11

[a] Met. is the magnetic cleaning method (AF, alternating field and Th, thermal), MAD is the maximum value of the mean angular deviation accepted for a sample, $S/n/N$ are the numbers of sites/accepted samples/measured samples, D/I is the declination/inclination of the remanent magnetization, α_{95} is the radius of the 95% cone of confidence, k is the best estimate of the Fisher precision parameter, P_{lat}/P_{lon} is the latitude/longitude of the virtual geomagnetic pole position and d_p/d_m are the semi-axes of the 95% confidence ellipsoid of the geomagnetic pole position.

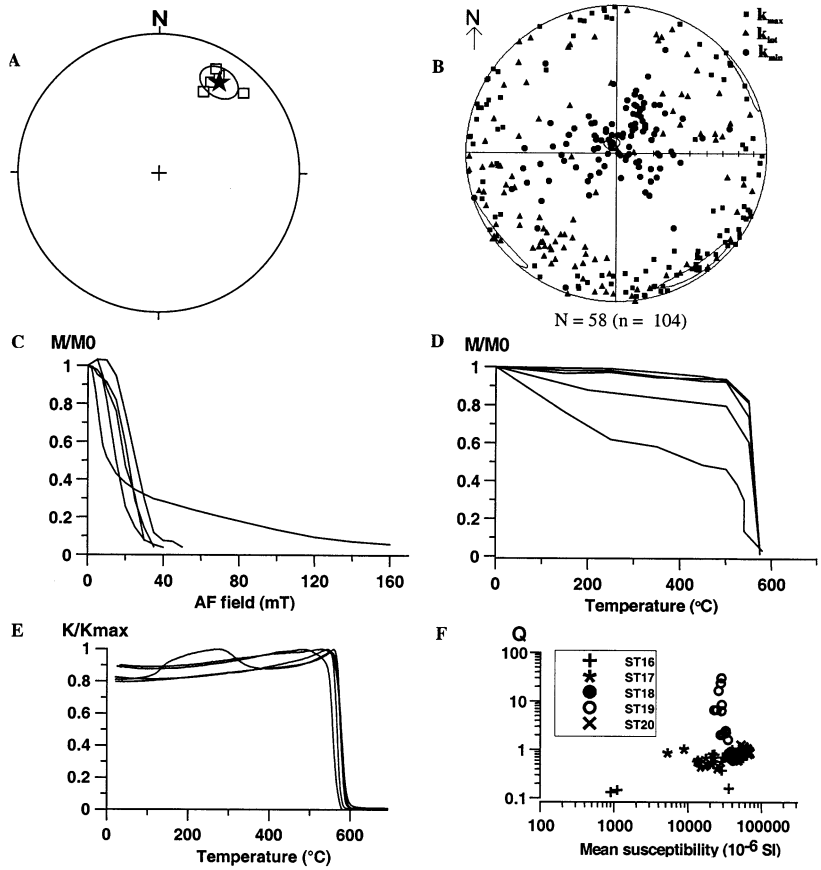


Fig. 9. Lower hemisphere equal area projection plot of the site mean remanence directions (A) and AMS directions of the sills from sites ST16–ST20 (B). The star in (A) shows the formation mean and the ring is the 95% cone of confidence of the mean. The ellipses in (B) are 95% confidence limits about each axis of the susceptibility ellipsoid. In (C)–(E) alternating field demagnetization, thermal demagnetization and heating curves of the susceptibility are shown for one sample from each of the dolerite sills, respectively. Log–log diagram of Q-values (Königsberger ratio) vs. the mean susceptibility is shown in (F).

6. Discussion and conclusions

6.1. The character of the SEDZ

In the Ljusdal batholith a shear zone that crosscuts the rock close to the locality KA06 (Fig. 3) defines the eastern boundary of the southern part of the Storsjön–Edsbyn deformation zone. East of site KA06 the rock has magnetic foliation planes with shallow dip towards the southeast and sub-horizontal lineations that strike E–W. This fabric is thought to represent a regional schistosity. To the west of this site (at sites KA05–KA12) the magnetic foliation planes dip steeply and show mainly N–S or NNW–SSE trending strike directions. Since there is no geological or AMS evidence that the Revsund type granite that was sampled west of the zone (sites KA13–KA16) has been subject to deformation caused by the SEDZ, the width of the zone in this area can be estimated to be 10–12 km.

Within the part of the Ljusdal batholith deformed by the SEDZ (sites KA05–KA12) two AMS patterns are found. NNW–SSE trending planes of magnetic foliation with maximum and intermediate anisotropy axes that create a girdle pattern define one type of fabric. This fabric has prolate and oblate shaped ellipsoids and is be-

lieved to be a result of both pure and simple shear deformations. Rather narrow zones of N–S trending planes of foliation cut this pattern. The latter zones have sub-vertical magnetic lineation and mainly prolate shaped anisotropy ellipsoids and may therefore have undergone a dominant pure shear deformation with dip-slip movement. The N–S trending zones are younger than the NNW–SSE pattern, but both fabrics appear to follow the regional trend of the SEDZ and may thus reflect two generations of deformation activity along the zone.

In the northern part of the SEDZ the magnetic fabric of the deformed Revsund granite (sites ST04–ST07) shows steeply dipping foliation and a rotation of the lineation from sub-horizontal to vertical when approaching the center of the SEDZ (sites ST08–ST10, ST16 and ST21) (Fig. 4). The pattern is consistent with pure-shear dominated transpression (e.g. Tikoff and Teyssier, 1994; Hrouda, 1994) probably caused by NNE–SSW compressive deformation. Tikoff and Teyssier (1994) show that the pure shear component develops faster than the simple shear component in this environment, and the tectonic fabric will therefore show a vertical lineation in highly deformed areas, whereas it will show more of the strike-slip component if the degree of deformation is lower. Our data therefore support the conclusion made by Bergman and Sjöström (1994) that the SEDZ is a transpression zone developed in a NNE–SSW compressive deformation. A change in the magnetic pattern is observed between the isotropic fabric at sites ST01–ST03 and the anisotropic fabric at sites ST04–ST07. This suggests that the rock at sites ST04–ST07 was affected by the

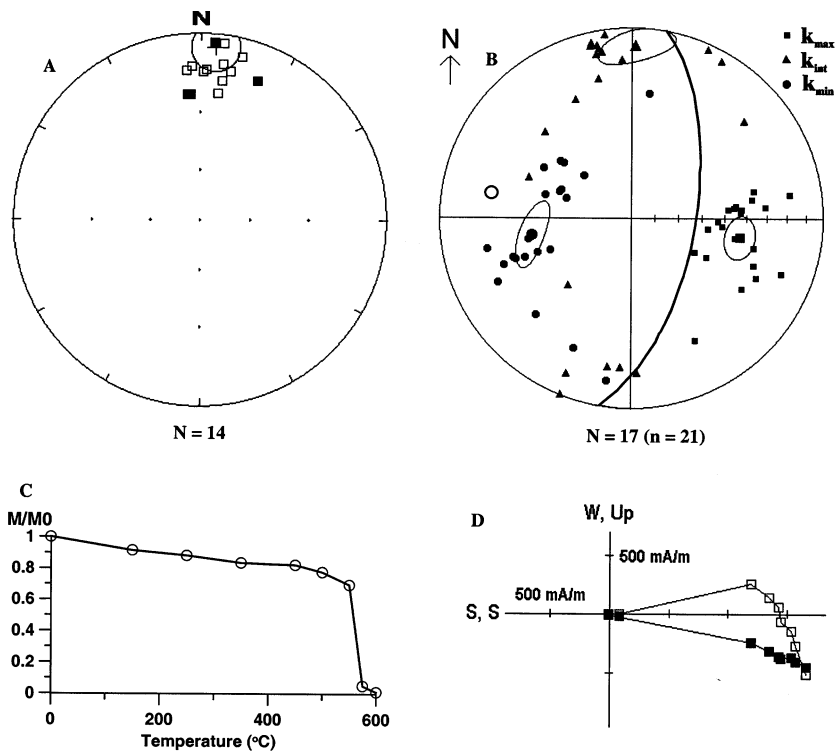


Fig. 10. Lower hemisphere equal area projection plot of the remanence directions (A) and AMS directions (B) of the dolerite dyke at site ST23. The ellipse in (A) shows the 95% cone of confidence of the mean. In (B) the plane of the dyke (great circle) and its pole (ring) are plotted. The ellipses are 95% confidence limits about each axis of the susceptibility ellipsoid. In (C) and (D) an example of a thermal demagnetization from 150 to 600°C and the corresponding vector plot is shown. Open (closed) symbols in (D) show vertical (horizontal) projection.

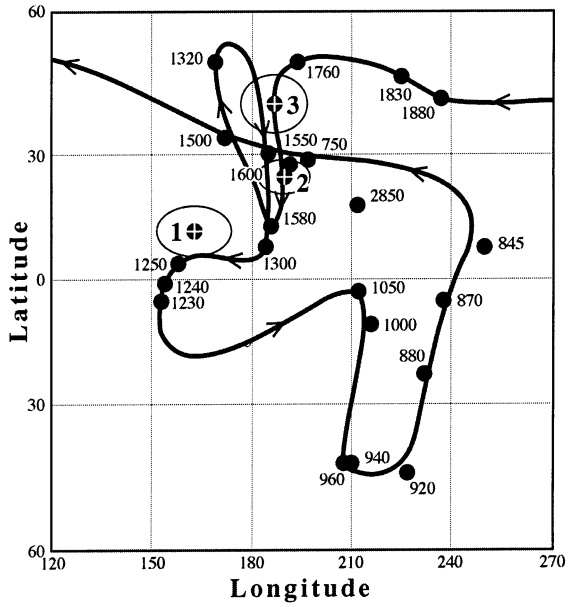


Fig. 11. The Apparent Polar Wander Path (APWP) for the Fennoscandian Shield (after Elming et al., 1993). Closed symbols along the APWP denote successive grand mean poles. Symbols with a white cross denote paleopoles from this study, shown with their circles of confidence. (1) Dolerite sill formation mean direction (mean of sites ST16–ST20, Table 1). (2) Dolerite dike (site ST23); the pole is calculated from the mean remanence direction defined from thermal demagnetization data (Table 1). (3) The Ljusdal batholith (site KA02); the pole is calculated from the mean remanence direction defined from AF demagnetization data (Table 1).

SEDZ, whereas at sites ST01–ST03 the rock still preserves a magmatic fabric. This change in fabric may define the eastern boundary of this part of the SEDZ, which then would be situated somewhere between the localities ST03 and ST04 (Fig. 4). The estimated width of the SEDZ based on AMS data is here 20–25 km.

The 1.70 Ga old Rätan granite (U–Pb, Delin, 1996) has an average degree of magnetic anisotropy of $P = 1.14$, which is higher than many sites inside the SEDZ. In the field, however the rock appears as almost isotropic and no microscopic evidence indicates plastic deformation of the granite. The relatively high P-value is most likely related to the high mean susceptibility of the rock (Fig. 7C). There is a sub-horizontal NW–SE trending strike direction of the magnetic lineation, that is consistent at all five sites (ST11–ST15, Fig. 4), and it may indicate a NW–SE directed magma flow (Bouillin et al., 1993). The change in dip of the magnetic foliation planes from sub-horizontal in the center to sub-vertical at the edge probably shows how the magnetic fabric is related to the shape of the pluton. The magnetic fabric of this granite is therefore believed to display a magmatic flow pattern of the final stages of the magma emplacement. It may indicate that the Rätan granite was emplaced in a NW–SE directed extensive environment, which contradicts a compressive environment as suggested by Bergman and Sjöström (1994). The emplacement of the granite took place in an early stage of the Gothian orogeny ($\sim$1.75–1.55 Ga, Gorbatschev and Bogdanova, 1993). The Gothian orogeny and the magmatism of the TIB are suggested to be related to subduction to the present west of the Baltic Shield (Gaál and Gorbatschev, 1987) and the TIB granites were emplaced in a spreading environment (Ahl et al., 1999; Åhäll and Larson, 2000), which is supported by our data.

6.2. Ages of deformation

The $\sim$1.84 Ga old (U–Pb, Delin, 1993) Ljusdal batholith is thought to belong to a late stage intrusion of subduction related magmatism during the Svecofennian orogeny, occurring mainly in the interval 1.9–1.86 Ga ago (Åhäll and Larson, 2000 and references therein). The timing of the regional metamorphism affecting the granodiorite is however still enigmatic. If the remanent magnetization of the Ljusdal batholith (decl. = 7° and incl. = 25°) is primary, a $\sim$30° clockwise rotation and a tilt of $\sim$10° would be required in order to change the direction from its expected direction (decl. = 335° and incl. = 35°), corresponding to the 1.84 Ga reference pole on the APWP for the Fennoscandian Shield (Fig. 11). A clockwise rotation is not in accordance with the dextral shear of the SEDZ and the dextral shear along the E–W trending margin of the northern edge of the batholith as suggested by Bergman and Sjöström (1994). A remagnetization therefore seems more probable. Pressure and temperature estimates

(Bergman and Sjöström, 1994) indicate that the metamorphism affecting the Ljusdal batholith was of amphibolite grade and therefore remagnetized the rock. It is thus possible that the defined paleomagnetic age of ~1.75–1.8 Ga is a metamorphic age and that there may be a causal relation between the metamorphism and the emplacement of the ~1.80 Ga old (Claesson and Lundqvist, 1995; Delin, 1996) Revsund granitoids. If the remanence was acquired during or after the metamorphism, the vector direction may have been significantly deflected from the direction of the ambient field at the time due to the high degree of anisotropy of magnetic susceptibility ($P = 1.5$). This degree of anisotropy could deflect TRM by as much as ~23° (Dunlop and Özdemir, 1997). Total deflection of the remanence vector in a specific rock is difficult to determine unless one knows the ambient field direction at the time and place of magnetization, and also the degree of anisotropy of the remanence carrier(s) (e.g. Lowrie and Hirt, 1986; Cogné, 1988).

The undeformed Postjotnian sill that truncates a plastic mylonite belonging to the SEDZ (site ST16, Fig. 4), gives a minimum age of the plastic deformation of the shear zone by ~1250 Ma. It also seems unlikely that the ~1.5–1.6 Ga dolerite dyke (site ST23), situated only ~0.5–1 km east of the zone center, would be unaltered if the zone had been active after the intrusion of the dyke, especially since the granodioritic host rock has a NW–SE trending foliation parallel to the SEDZ. The paleomagnetic data from this dyke therefore indirectly show that the SEDZ is most likely older than ~1.5–1.6 Ga.

The 1.80 Ga age of the SEDZ presented by Högdahl et al. (1996) probably dates one deformation event, but AMS and aeromagnetic data indicate the possibility of multiple generations of deformation. The development of a major shear zone such as the SEDZ (>200 km in length and 10–25 km wide) most likely occurred over a period of millions of years. The age of the titanites is approximately coeval with the ~1.80 Ga old Revsund granitoid (Claesson and Lundqvist, 1995) but the Revsund granite is deformed by the SEDZ (this study) and this indicates deformation along the SEDZ after 1.80 Ga ago. Early TIB-magmatism (1.81–1.77 Ga) is believed to have occurred in relation with eastward subduction (Ahl et al., 1999; Åhäll and Larson, 2000), but whether the tectonic conditions were governed by extension (favored by Åhäll and Larson, 2000) or compression (favored by Ahl et al., 1999) is still debated. The compressive deformation along the SEDZ indicated in this study support Ahl et al. (1999), however our data indicate compression more from the present SSW than west. Northward compression at ~1.80 Ga is reported in southwestern Finland (Lindroos et al., 1996), which together with our data indicate the possible existence of a regional northward compressive stress field at that time. The timing of early TIB-magmatism (1.81–1.77 Ga as indicated by Åhäll and Larson, 2000; Ahl et al., 1999) and of the SEDZ (~1.80 Ga, Högdahl et al., 1996; and possibly younger, this study), suggests an overlap of ~10 m.y. between northward compressive deformation and eastward subduction. To fit the early TIB-magmatism with a northward compression Åhäll and Larson (2000) suggest a change in the stress conditions at 1.81 Ga, which however is ~10 m.y. to early according to the existing radiometric dating (Högdahl et al., 1996). The geological evolution in the beginning of the Gothian orogeny (1.75–1.70 Ga, Gaál and Gorbatschev, 1987) is still enigmatic, and the period between 1.80 and 1.75 Ga ago may represent an initial stage of this orogeny. It is therefore possible that the compressive stress field from the present SSW indicated in this study is related to an initial stage of plate collision. This collision between the southwestern Scandinavian subprovince and older parts of the Fennoscandian Shield resulted in a system of regional NW–SE trending intra-plate deformation zones, one of which is the SEDZ (Fig. 12). At this time, or somewhat later, the Fennoscandian Shield is also amalgamated to the Ukrainian Shield (Elming et al., in press) possibly forming a Pre-Rodinian supercontinent.

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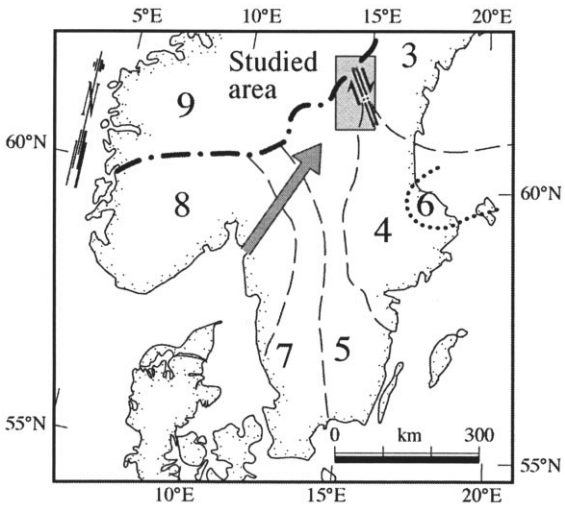


Fig. 12. Map showing the southern part of the Fennoscandian Shield with the same block division as in Fig. 1. The gray arrow indicates the compressive force inferred from AMS data and black arrows show the dextral shear along the SEDZ. Parallel hatched lines indicate the SEDZ.

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